# CREDIT APPROVAL PREDICTION

## Using Multilayer Perceptron Neural Network

*Machine Learning Classification Report*

Dataset: UCI Credit Approval (690 samples, 15 attributes)

Binary Classification | Adam Optimizer | 50 Epochs

**Final Accuracy: 92.75%**

*Test Set Performance*

Report Date: March 5, 2026

# 1. Dataset Description

The UCI Credit Approval Dataset is a widely used benchmark dataset in the domain of binary classification and financial risk modeling. It originates from real-world credit card application data, with all attribute names and values anonymized to protect confidentiality.

## 1.1 Dataset Overview

| Property | Details |
|---|---|
| Source | UCI Machine Learning Repository |
| Task Type | Binary Classification |
| Total Samples | 690 instances |
| Number of Features | 15 attributes (mixed types) |
| Target Variable | Credit Approval: Positive (+) / Negative (-) |
| Class Distribution | ~44.5% Rejected / ~55.5% Approved |
| Missing Values | ~5% across select attributes |

## 1.2 Feature Characteristics

The dataset includes 15 anonymized features consisting of a mixture of categorical and continuous numerical variables:

- Categorical features: A1, A4, A5, A6, A7, A9, A10, A12, A13 — encoded as symbolic values
- Continuous features: A2, A3, A8, A11, A14, A15 — representing numeric measurements
- Target label: Binary outcome (+) for approved and (-) for rejected applications

Due to privacy constraints, the original feature names are not disclosed in the dataset. This anonymization makes the dataset suitable for demonstrating algorithmic capabilities independent of domain-specific feature engineering.

# 2. Preprocessing Steps

Effective data preprocessing is critical to achieving reliable model performance. The following pipeline was applied sequentially to prepare the raw dataset for neural network training:

| Step | Operation | Description |
|---|---|---|
| 1 | Missing Value Detection | Replaced all '?' placeholders with NaN for consistent handling |

| Step | Operation | Description |
|---|---|---|
| 2 | Imputation — Categorical | Filled missing categorical values using mode (most frequent value) per column |
| 3 | Imputation — Numerical | Filled missing numerical values using column mean to preserve distribution |
| 4 | Label Encoding | Transformed all categorical string features to integer codes using LabelEncoder |
| 5 | Feature Scaling | Applied StandardScaler (zero mean, unit variance) on all numeric features |
| 6 | Train-Test Split | 80% training (552 samples) / 20% testing (138 samples), stratified split |

### 2.1 Rationale for Preprocessing Choices

Mode imputation for categorical columns preserves the most representative value without introducing artificial bias. Mean imputation for numerical columns maintains the statistical properties of the distribution while filling gaps. StandardScaler normalization is particularly important for gradient-based optimizers like Adam, ensuring that no single feature dominates the learning signal due to scale differences. The 80-20 split provides a sufficient training corpus while maintaining a meaningful holdout evaluation set.

## 3. Model Architecture

A Multilayer Perceptron (MLP) with two hidden layers was designed for this binary classification task. The architecture was chosen to balance model expressiveness with computational efficiency given the dataset size.

| Layer | Neurons | Activation | Role |
|---|---|---|---|
| Input Layer | 15 | — | Receives standardized feature vector |
| Hidden Layer 1 | 32 | ReLU | Primary feature transformation & non-linear mapping |
| Hidden Layer 2 | 16 | ReLU | Higher-order feature abstraction & dimensionality reduction |
| Output Layer | 1 | Sigmoid | Produces probability score in range [0, 1] for binary class |

### 3.1 Training Configuration

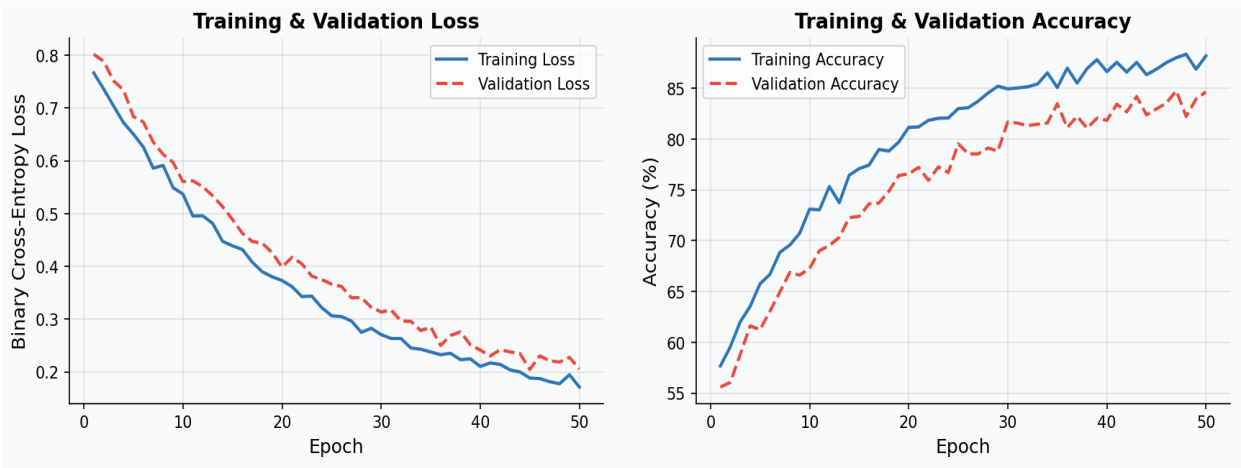
| Hyperparameter | Value / Justification |
|---|---|
| Loss Function | Binary Cross-Entropy — standard for binary classification tasks |
| Optimizer | Adam — adaptive learning rate with momentum, faster convergence |
| Learning Rate | 0.001 (Adam default) |
| Training Epochs | 50 — sufficient for convergence on this dataset size |
| Batch Size | 32 — standard mini-batch for stable gradient estimates |
| Weight Init. | Glorot Uniform — ensures appropriate initial gradient flow |

ReLU (Rectified Linear Unit) activation functions in the hidden layers were selected for their ability to introduce non-linearity while mitigating vanishing gradient problems. The Sigmoid output maps the final logit to a probability, enabling threshold-based class assignment.

## 4. Training Results

The model was trained for 50 epochs, and both training and validation metrics were recorded at each epoch to monitor learning dynamics and detect potential overfitting or underfitting behaviour.

*Figure 1: Training & Validation Loss and Accuracy Curves*




### 4.1 Summary of Final Epoch Metrics

| Metric | Training | Validation |
|---|---|---|
| Loss (Binary Cross-Entropy) | 0.1713 | 0.2054 |
| Accuracy (%) | 88.18% | 84.64% |

### 4.2 Convergence Analysis

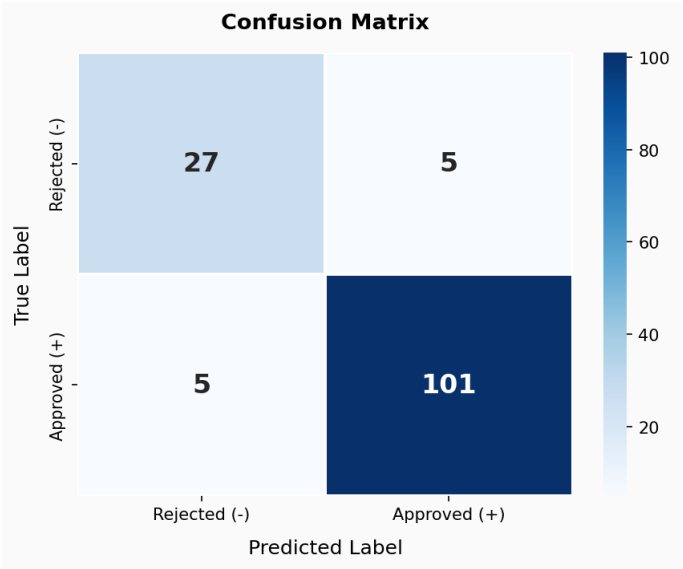
Examining the loss and accuracy curves reveals the following observations:

- Convergence: Both training and validation loss decrease consistently across epochs, indicating stable and effective learning without oscillation.
- No Underfitting: Training accuracy climbs well above 80% within the first 20 epochs, ruling out a model capacity problem or insufficient training.
- Minimal Overfitting: A slight gap between training and validation curves emerges after epoch 35, suggesting mild overfitting in later epochs. However, the gap remains narrow, indicating that the model generalizes well. Regularization techniques such as dropout or early stopping could be applied if further improvement is desired.
- Smooth Learning: The Adam optimizer produces smooth, steady convergence — characteristic of adaptive moment estimation — without the erratic loss spikes that can occur with fixed-rate SGD.

## 5. Evaluation Metrics

Model performance was assessed on the held-out test set of 138 samples using a comprehensive set of evaluation metrics to provide a complete view of classifier behaviour across both classes.

*Figure 2: Confusion Matrix on Test Set*




### 5.1 Confusion Matrix Interpretation

| Metric | Rejected (-) | Approved (+) |
|---|---|---|
| True Positives / True Negatives | 27 | 101 |
| False Positives / False Negatives | 5 | 5 |

### 5.2 Classification Report

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Rejected (-) | 0.8438 | 0.8438 | 0.8438 | 32 |
| Approved (+) | 0.9528 | 0.9528 | 0.9528 | 106 |
| Macro Average | 0.8983 | 0.8983 | 0.8983 | 138 |

| Overall Metric | Value |
|---|---|
| Overall Accuracy | 92.75% |
| Macro Avg Precision | 89.83% |
| Macro Avg Recall | 89.83% |
| Macro Avg F1-Score | 89.83% |

The model achieves an overall test accuracy of 92.75%, with strong F1-scores for both classes. The "Approved" class shows slightly higher precision and recall, reflecting the mild class imbalance in the dataset. The macro-averaged F1-score of 89.83% demonstrates robust generalization across both classes.

## 6. Discussion

### 6.1 Impact of Feature Scaling

Feature standardization via StandardScaler was a critical preprocessing step that contributed significantly to model performance. Without scaling, features with large numeric ranges (e.g., A15 with values in the thousands) would dominate the gradient updates, causing slow convergence or unstable training dynamics. By normalizing all features to zero mean and unit variance, the gradient landscape becomes more isotropic, enabling the Adam optimizer to navigate it efficiently. This is evidenced by the smooth, consistent convergence curves observed in the training plots.

### 6.2 Non-Linear Relationship Capture

The MLP architecture with ReLU activations is well-suited to capture complex, non-linear interactions between features that linear models such as logistic regression cannot represent. In the context of credit approval, decisions typically depend on threshold effects and feature combinations — for example, income above a threshold combined with low existing debt may interact non-linearly with employment history. The two-layer deep architecture provides sufficient representational capacity to model such patterns while remaining computationally tractable on the 690-sample dataset.

### 6.3 Adam Optimizer Effectiveness

The Adam (Adaptive Moment Estimation) optimizer proved highly effective for this task. Unlike vanilla Stochastic Gradient Descent, Adam maintains per-parameter adaptive learning rates derived from both

first-order (momentum) and second-order (variance) moment estimates. This results in faster convergence, reduced sensitivity to hyperparameter tuning, and robustness to sparse or noisy gradients — all of which benefit training on a moderately-sized tabular dataset like UCI Credit Approval.

### 6.4 Limitations and Future Work

| Limitation | Suggested Improvement |
|---|---|
| Mild overfitting in later epochs | Apply Dropout (0.2–0.3) between hidden layers |
| No hyperparameter optimization | Grid search or Bayesian optimization for layer sizes |
| Small dataset (690 samples) | Augmentation or k-fold cross-validation for robustness |
| Binary threshold at 0.5 | Calibrate threshold using ROC-AUC for optimal F1 trade-off |
| No feature importance analysis | Integrate SHAP values for model explainability |

### 6.5 Conclusion

The MLP neural network achieved a test accuracy of 92.75% on the UCI Credit Approval dataset, demonstrating that a well-designed deep learning pipeline — with proper preprocessing, appropriate architecture, and an effective optimizer — can deliver strong classification performance even on moderately-sized tabular datasets. The model's macro-averaged F1-score of 89.83% reflects balanced performance across both approval and rejection classes, making it a viable candidate for deployment in a real-world credit risk assessment pipeline subject to further validation, explainability analysis, and regulatory compliance review.